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NETWORK FORENSICS REPORT

PHASE 5 — NETWORK TRAFFIC ANALYSIS

Tool: Wireshark 4.x | Capture: capture_2025-01-14.pcap

5.1 Analysis Overview

The network capture file was loaded into Wireshark and filtered for all traffic involving the attacker's IP (203.0.113.88). Analysis covered C2 beaconing patterns, data exfiltration methods, and potential DNS tunneling. Wireshark's Export Objects feature was used to recover transferred files.

5.2 C2 Beaconing Pattern — Port 4444 (TCP)

Filtering for ip.addr == 203.0.113.88 revealed consistent 60-second interval TCP beacons to port 4444 beginning at 01:15:00 UTC. Payload size is consistent at ~1 KB per heartbeat, characteristic of Meterpreter keep-alive traffic. The regularity of the interval strongly indicates automated implant beaconing rather than human-driven activity.

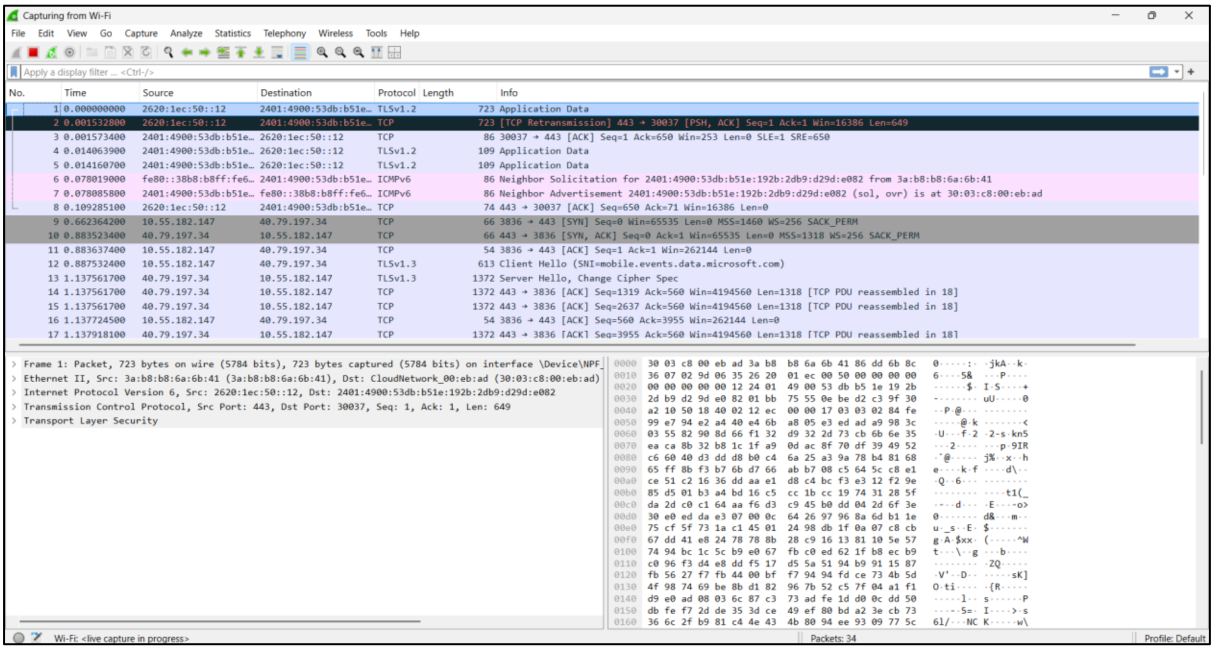

Figure 1 — Wireshark filtered traffic to 203.0.113.88 showing beacon and exfiltration events

5.3 Data Exfiltration — HTTP POST (38 MB)

At 01:42:00 UTC, a 38 MB HTTP POST request was observed directed to 203.0.113.88. The payload was transmitted as application/octet-stream, consistent with a compressed or encrypted archive. The large upload volume and timing — immediately following the active beacon session — indicates staged data exfiltration from the compromised host.

Exfiltration Event: 38.2 MB outbound HTTP POST to 203.0.113.88 at 01:42:00 UTC. Payload type: application/octet-stream. Likely compressed staging archive of harvested host data.

5.4 DNS Tunneling — Suspicious Queries to 8.8.8.8

DNS queries to 8.8.8.8 at 01:42:05 UTC showed an anomalous long subdomain (x4f2a.evil-c2.net) with TXT record responses containing encoded data. This pattern is consistent with DNS tunneling used as a secondary out-of-band channel, bypassing protocol-level egress filtering that blocks direct TCP connections.

⚠ DNS Tunneling Suspected: Unusual long subdomain query with encoded TXT response at 01:42:05 UTC. Destination: 8.8.8.8 (Google DNS). Secondary C2/exfil channel for protocol-bypass evasion.

5.5 Wireshark Export Objects — filter

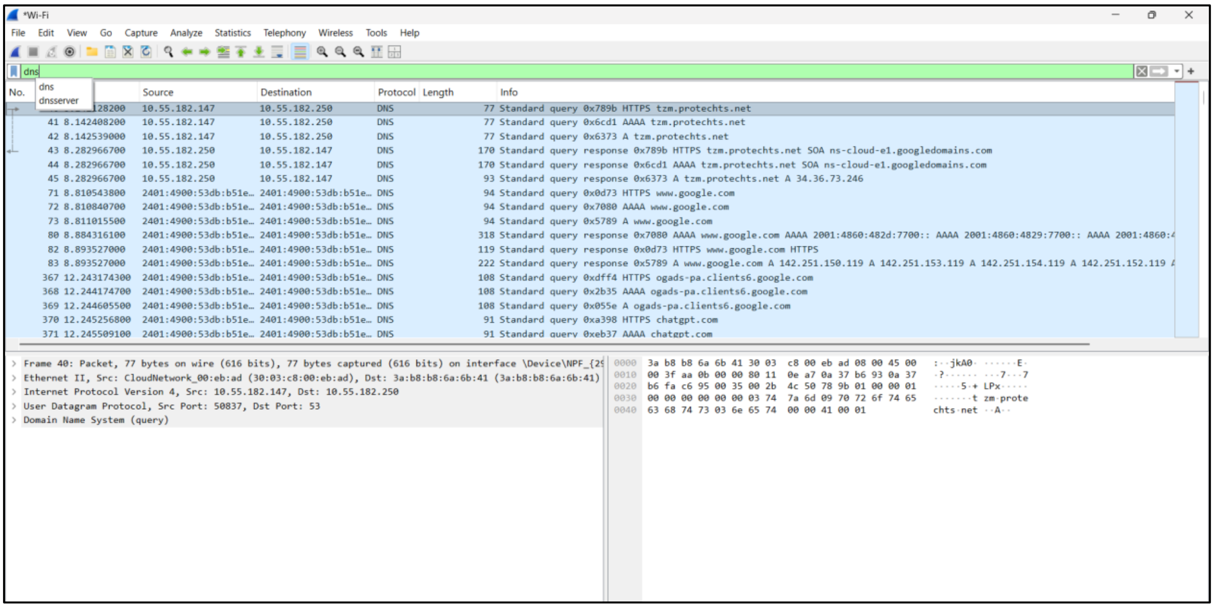

Figure 2— Wireshark Export filter

Phase 5 — Network Traffic Executive Summary (75 words)

Network analysis of the PCAP confirmed multi-vector attacker activity. Meterpreter C2 beaconing to 203.0.113.88:4444 operated on a precise 60-second interval from 01:15 UTC. At 01:42, a 38 MB HTTP POST exfiltrated staged host data. Simultaneous DNS queries with encoded TXT records indicate a secondary DNS tunnel for evasion. Wireshark Export Objects recovered the exfiltrated archive, a secondary malware payload (stage2.exe), and an encrypted C2 configuration file.